



Effectiveness of Privacy-preserving Algorithms in LLMs: A Benchmark and Empirical Analysis

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Abstract

Preserving individual privacy is crucial when interacting with Large Language Models (LLMs) during both training and inference stages. Privacy leakage at either stage can lead to irreversible negative consequences. Although data-level privacy-preserving algorithms have been developed for smaller Natural Language Processing (NLP) models, their application to LLMs has not been extensively explored. Moreover, with plenty of algorithms emerging, it brings challenges for organizations or researchers to compare and evaluate these different algorithms to select the most suitable one for their certain requirements. To address these challenges, we introduce "Privacy-preserving4LLM Benchmarking", a systematic evaluation framework that systematically assesses different privacy-preserving algorithms' utility-privacy trade-offs across different LLM architectures. Our framework evaluates these algorithms in three practical scenarios: protecting training data only, user queries only, and both. We also introduce a novel Parameter Optimizer to ensure fair comparisons. To quantify privacy protection levels, we use exposure metrics, where canary data sequences are intentionally inserted into training data to measure information memorization and potential leakage. Our study presents a comprehensive empirical analysis comparing three privacy-preserving algorithms across three LLM architectures (Mistral-7B, Llama2-7b, Falcon-7b) using three different datasets. Our findings reveal that algorithm selection, protection scenarios, LLM architectures, and privacy budget settings all impact the utility and privacy level.

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CCS Concepts

• **Security and privacy** → **Data anonymization and sanitization**; • **Computing methodologies** → *Natural language generation*.

Keywords

Large Language Models; Privacy-preserving; Differential Privacy; Algorithms; Analysis

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1 Introduction

With the increasing capabilities of large language models (LLMs), a growing number of industries are beginning to adopt and integrate them into their workflows, spanning sectors such as code generation [35], education [17], healthcare [31], customer service [25], and finance [40]. While these models significantly enhance operational efficiency and service delivery, they raise critical privacy and data security concerns. These concerns become particularly significant when fine-tuning pre-trained LLMs for specific domains. This process requires domain-specific training data that might contain personally identifiable information (PII), sensitive records, and proprietary knowledge. A fundamental characteristic of LLMs is their tendency to memorize training data [15] - storing and reproducing specific phrases or passages [28]. While this memorization capability enhances model performance, it also creates vulnerabilities to various privacy and security attacks [1, 2, 32, 44]. Recent research has exposed multiple attack vectors that exploit these vulnerabilities, including jailbreak attacks [20, 38] and training data extraction attack [4]. Through these attacks, models may unintentionally disclose sensitive information from their training data [28]. The privacy risks extend beyond the training phase to the inference stage, where LLMs can deduce personal attributes from user inputs [30], creating potential privacy leakage in training and query data.

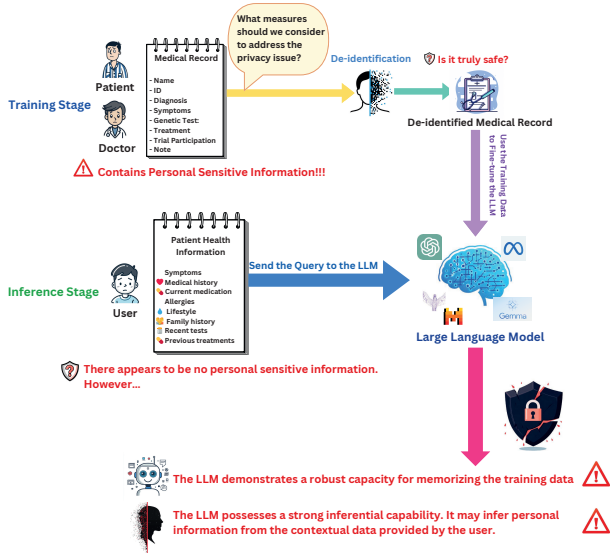


Figure 1: An overview of privacy vulnerabilities in LLM systems: At the training stage, even after de-identification, LLMs can still memorize the remaining information and potentially infer sensitive details through their strong reasoning capabilities. At the inference stage, while user queries may appear to contain no explicit personal information, the contextual understanding of LLMs could lead to unintended exposure of sensitive information.

Deleting or correcting sensitive data used to train the LLMs to eliminate privacy leakage can be extremely challenging due to the LLMs' lack of transparency and limited interpretability [39]. To avoid irreparable loss due to leakage, customers may refuse to share their personal data to be used in training these models. Many companies have also implemented stringent policies, including restricting the use of platforms like ChatGPT among employees, to prevent the exposure of sensitive data¹. While aimed at protecting privacy, such restrictions may hinder the progressive development and perfection of LLMs. Therefore, it is essential to find solutions that effectively address privacy concerns while maintaining the utility of these models to the greatest extent possible.

Recently, several data-level privacy-preserving algorithms [5, 8, 11, 14, 41–43] based on validated and widely recognized algorithms such as differential privacy, have been proposed. These algorithms are particularly promising because they can be implemented during both the training and inference stages. While these algorithms have been verified for their effectiveness on traditional NLP models with millions of parameters, their potential application to LLMs with billions of parameters and different architectures remains largely unexplored. This gap is significant because LLMs' enhanced capacity for memorization and information processing introduces novel privacy concerns that require careful examination. Current challenges lie in the absence of a systematic evaluation framework, which makes it difficult for researchers and organizations to select appropriate privacy solutions that meet their specific needs. These promising privacy-preserving algorithms thus remain largely theoretical, despite their potential to address critical privacy challenges in LLM systems. To address these limitations, we propose the **first**

comprehensive framework, "Privacy-preserving4LLM Benchmarking", to evaluate privacy-preserving algorithms in LLMs. This work makes four **major contributions**:

1. Our novel benchmarking framework provides the first systematic evaluation methodology for assessing privacy-preserving algorithms in LLM contexts, enabling comprehensive performance comparisons across different LLM architectures and diverse datasets.
2. The framework enables assessment under three real-world scenarios: protecting training data only, query data only, and both.
3. The framework incorporates a Parameter Optimizer that ensures a fair comparison of different privacy-preserving algorithms, offering a standardized evaluation methodology.
4. Through extensive empirical studies, we conduct the first comprehensive evaluation comparing three privacy-preserving algorithms (SanText+, CusText+, and Mahalanobis mechanism) across three LLM architectures (Mistral-7B, Llama2-7b, Falcon-7b) and diverse datasets, establishing reproducible baseline performance metrics for future research in this area.

Our experimental results reveal three **key findings**: (1) CusText+ generally achieves superior utility, particularly under stringent privacy requirements ($\epsilon = 1$), while SanText+ achieves similar performance with CusText+ under relaxed privacy settings. The Mahalanobis mechanism provides complete privacy protection at a significant cost to utility performance. (2) Protecting query only (Scenario 2) or both training and query data (Scenario 3) leads to notably reduced utility compared to protecting training data alone (Scenario 1). (3) Llama2-7b shows the best compatibility with our privacy protection methods, consistently maintaining good performance. Mistral-7B exhibits robust inherent privacy safeguards, requiring additional protection only when handling frequently repeated sensitive information in the training data. In contrast, Falcon-7b shows increased vulnerability even with protection.

2 Related Work

• **Privacy Threats:** Recent studies have shown that LLMs can **memorize** specific details from their training data, including personal information, URLs, and code snippets [4]. This memorization capacity increases with larger model sizes, leading to privacy leakage. Nasr et al. [24] identified memorized content in trillions of tokens across terabyte-scale datasets generated by multiple models. Another significant concern is **jailbreaking** - the potential for LLMs to be manipulated through carefully crafted prompts to generate unsafe content or reveal private information, despite the intentions of their creators [6, 19, 20, 22, 26, 38]. Privacy risks extend beyond training data to the **inference process** itself. Staab et al. [30] showed that malicious actors can infer private information from a user's online posts by feeding them to pre-trained LLMs.

• **Privacy-Preserving Techniques:** Several frameworks have been proposed to protect user privacy **during LLM inference stage**, including Hide and Seek (HaS) [9], "Life of PII" [10], and Privacy-Preserving via Text Sanitization (PP-TS) [18]. These approaches employ techniques such as anonymization, obfuscation, and sanitization of PII. However, they remain vulnerable to inference attacks due to LLMs' robust contextual understanding capabilities. To address **training data leakage**, Vats et al. [36] proposed using LLMs to generate substitutes for masked tokens. While their

¹<https://www.hr-brew.com/stories/2023/05/11/these-companies-have-banned-chatgpt-in-the-office>

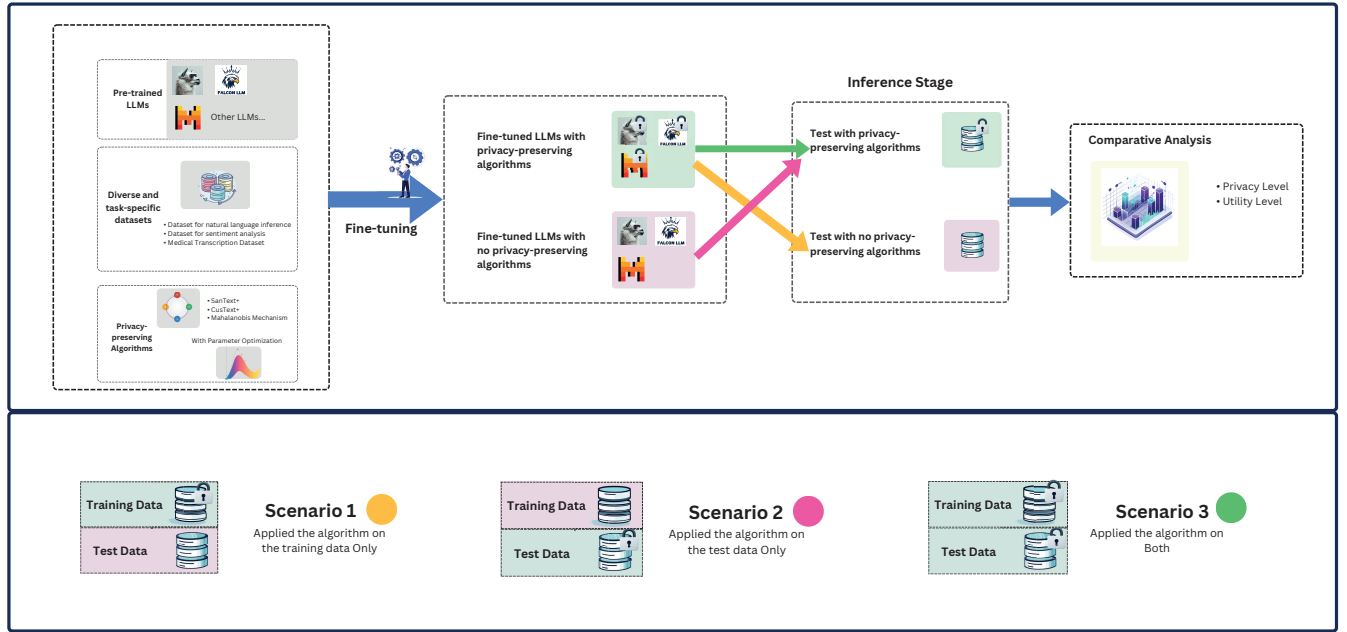


Figure 2: Proposed Privacy-preserving4LLM Benchmarking Framework

approaches maintain utility, they lack rigorous privacy guarantees. **Differential privacy**, introduced by Dwork [13], provides a rigorous framework for protecting data privacy. It has been widely recognized as a strong and mathematically rigorous approach to privacy protection, offering superior guarantees.

• **Existing Privacy Evaluation Frameworks:** Several researchers have explored frameworks and benchmarks for privacy protection in large language models. Ullah et al. [33] presented PrivChat-GPT, which combines multiple privacy techniques. While theoretically promising, this framework awaits practical implementation and testing. Plant et al. [27] proposed a benchmark for evaluating privacy-preserving algorithms on a large, multi-lingual sentiment analysis dataset. Li et al. [21] introduced RAPT, a framework that addresses privacy concerns throughout both training and query processes. RAPT employs text transformation techniques to ensure local differential privacy before data reaches the service provider’s learning system, though its applications remain relatively narrow in scope. Most relevant to our work is Meisenbacher et al. [23]’s analysis of privacy protection methods in language models. While they evaluated seven different approaches, their study focused only on LSTM models and basic word-level metrics. Their work did not address broader concerns such as model memory effects or real-world usage scenarios where training and testing data require separate privacy handling.

3 Privacy-preserving4LLM Benchmarking Framework

We present Privacy-preserving4LLM, a benchmark framework for evaluating privacy-preserving algorithms across various LLM architectures and datasets. As illustrated in Figure 2, our framework introduces two key innovations. The first is a **Protection Scenario**

Manager that offers three selectable scenarios: protecting training data, protecting query data, or protecting both. While organizations can select scenarios matching their specific needs, our work covers all three scenarios to provide a complete understanding. The second innovation is a **Parameter Optimizer** that ensures fair comparisons between different privacy-preserving algorithms.

Our framework follows a structured process for each combination of LLM architecture and privacy-preserving algorithm. First, the Parameter Optimizer determines the optimal parameter for the privacy-preserving algorithm. These optimized algorithms are then applied to the data, followed by the Protection Scenario Manager determines and implements appropriate evaluation scenarios based on organizational requirements. Finally, we conduct comprehensive assessments of both privacy protection levels and utility (measured as accuracy) across different algorithms and architectures.

3.1 Privacy-Level Evaluation: Canary Exposure

In this study, we define privacy in LLMs as protecting sensitive training data from unintended memorization and exposure. We adopt the **exposure metric** introduced by Carlini et al. [3] to quantify how much sensitive information is memorized by LLMs. We represent sensitive information through ‘**canary data**’ – specific sequences intentionally inserted into the training data to serve as identifiable markers of information memorization. The **key objective** of this approach is to measure and quantify the risk that rare or unique training sequences could be **extracted** by potential attackers through targeted queries. We define our canary data using the format ‘patient name _____’, where the blank represents a position for random name sequences. In our experiments, we insert ‘**patient name elsa**’ as the canary sequence, with ‘elsa’ being the random component, with 26^4 possible permutations. The exposure

of this sequence is calculated as:

$$\text{Exposure} = \log_2(\text{Total number of guesses}) - \log_2(\text{Rank of canary})$$

For the sequence "elsa", "Total number of guesses" = 26^4 . This metric ranges from 0 to $\log_2(26^4)$, with higher values indicating stronger memorization. Lower exposure values suggest better privacy preservation. In this study, we also set an exposure threshold. This threshold represents the acceptable level of privacy risk (upper bound of acceptable exposure value) and can be customized by organizations based on their specific requirements. If the exposure value exceeds this threshold, the canary word is considered at risk of being exposed. The exposure threshold is calculated as: $\log_2(26^4) - \log_2(10)$. This threshold is based on our inference stage configuration, where the LLM samples one token from the top-10 most probable tokens according to their probability distribution. These exposure settings remain consistent throughout all privacy-level assessments in this study. In this paper, we use these consistent terms: **canary data** (inserted sensitive information), **exposure metric** (quantitative measure of memorization), **exposure threshold** (upper bound of acceptable exposure value), **extraction** (process of recovering canary data from trained models), and **be exposed** (when the exposure value exceeds the exposure threshold).

3.2 Protection Scenario Manager

The Protection Scenario Manager enables organizations to evaluate privacy-preserving algorithms under different protection contexts. As shown in Figure 2, our framework includes three scenarios that reflect different real-world privacy requirements: **Scenario 1: Training Data Protection Only** addresses cases where only training data contains sensitive information, while inference queries contain non-sensitive data. This scenario is relevant for organizations that release public models while protecting their proprietary or sensitive training data. **Scenario 2: Query Data Protection Only** represents a more practical setting where only inference queries contain sensitive information while training data includes non-sensitive content. This scenario reflects real-world applications where organizations can verify the safety of training data but must protect user privacy during model interactions. **Scenario 3: End-to-End Protection** implements the most stringent privacy requirements, protecting both training and inference data. This scenario is particularly relevant for high-sensitivity domains, such as healthcare, where both training data (e.g., patient records) and user queries (e.g., symptom descriptions) may contain sensitive information requiring protection. For clarity of terminology throughout this paper, **'test data'** refers to experimental evaluation data that simulates **'query data'** – real user inputs in deployed LLM systems.

3.3 Parameter Optimizer

Our review of the algorithms reveals that their utility and privacy levels depend not only on privacy budget (ϵ) – a parameter that controls privacy strength, where smaller values provide stronger protection but lower utility – but also on various other parameters. Simply using default parameter values may lead to suboptimal results. To identify the optimal parameters for each algorithm, ensuring fair comparison among privacy-preserving algorithms, we introduce a systematic Parameter Optimizer while maintaining a

fixed ϵ . Our approach first evaluates each algorithm's utility across different parameter configurations. To assess privacy protection, we insert canary data into the training set and measure the resulting exposure values. Among all parameter values, we select the one that maximizes utility while maintaining the exposure value below the exposure threshold.

3.4 Privacy-preserving Algorithms

In this section, we have a brief review of three different algorithms involved in this study. They are all **differential-privacy based**.

• **SANTEXT+** [43] is a text sanitization algorithm that replaces tokens with semantically similar alternatives, building on their novel privacy notion called "Utility-optimized Metric Local Differential Privacy". It enables distinguishing between sensitive and non-sensitive tokens, allowing common words to remain unchanged while focusing protection on potentially sensitive information.

• **CUSTEXT+** [8] is a customized text sanitization mechanism that builds on ϵ -differential privacy (DP) to protect sensitive information. The method uses two key components: a customized mapping scheme that creates smaller, semantically relevant output sets for each input token, and the Exponential Mechanism for token sampling. Moreover, it can preserve stopwords in their original form, which improves text utility without compromising privacy.

• **Mahalabobis Mechanism** [42] is a privacy-preserving text perturbation method that adds elliptical noise to word embeddings in the Euclidean space, with the noise scale calibrated by the regularized Mahalanobis norm. The method adds elliptical noise to word embeddings according to the covariance structure in the embedding space, stretching the noise equidistant contour in the direction that explains the largest variability in the word embedding vectors. This increases the substitution probability of words in the sparse region without sacrificing the overall utility.

4 Experiments and Results

4.1 Datasets

• **MTSamples**² includes sample transcription keywords from various medical specialties. Our objective in using this data set is to infer medical specialties from given transcription keywords. To clean and standardize the dataset, we removed under-represented and overlapping medical specialties and then normalized each sample to contain exactly 20 representative keywords. After preprocessing, our dataset contained 1,532 samples.

• **SST-2**[29] is the dataset utilised for conducting binary sentiment classification tests on sentences.

• **QNLI**[37] contains pairs of questions and paragraphs taken from Wikipedia. The objective is to determine whether the given context sentence contains the answer to the posed question.

4.2 Experimental Setup

To conduct our experiments, we used Google Colab with an A100 GPU and Katana[34]. For our analysis, we selected three widely-adopted 7-billion parameter language models available on the Hugging Face platform³: Llama2-7b, Falcon-7b, and Mistral-7B. In order

²<https://www.kaggle.com/datasets/tboyle10/medicaltranscriptions>

³<https://huggingface.co/meta-llama/Llama-2-7b-hf>; <https://huggingface.co/tiiuae/falcon-7b>; <https://huggingface.co/mistralai/Mistral-7B-Instruct-v0.3>

to efficiently fine-tune the LLMs, we employed Low-Rank Adaptation (LoRA) [16], a training approach that substantially reduces the number of trainable parameters while maintaining model performance. The hyperparameters used for training and testing are detailed in Appendix C.

4.3 Parameter Optimization

In this section, we demonstrate our parameter optimization process for each privacy-preserving algorithm. For these experiments, we use the MTSamples dataset, maintain $\epsilon = 3$, and employ GloVe-300d embeddings across all algorithms. To ensure consistency in our comparative analysis, we perform the optimization using Llama2-7b (epochs = 120; number of insertions = 200) as our reference model and apply the resulting parameters across all three LLMs in our study. While organizations implementing these algorithms in practice should optimize parameters for their specific LLM architecture and use case, this standardized approach enables us to conduct a systematic comparison of the algorithms' relative performance.

• **SANTEXT+:** We first identify the parameter p as the key factor influencing both the utility and the privacy level of SANTEXT+. The parameter p is responsible for modifying the probability of non-sensitive words being sanitized. Our experimental results, shown in Figure 3, demonstrate that as the p value increases, the utility of the algorithm decreases. At the same time, the exposure of canary data also decreases with increasing p values. The most significant finding is that when p is lower than 0.4, the exposure of canary data exceeds the exposure threshold, indicating that canary data is at risk of being exposed. Consequently, we selected $p = 0.4$.

• **CUSTEXT+:** In the article by Chen et al. [8], we identified the parameter Top-K, which determines the number of semantically similar words used by the CUSTEXT+ algorithm to substitute each word. The CUSTEXT+ algorithm replaces each word with one of the k-most semantically similar words. Consequently, an increase in Top-k will inversely affect accuracy. To determine the optimal parameter value, we conducted experiments following the steps discussed previously, adjusting the Top-K parameter from 10 to 50. Our experimental results, shown in Figure 4, confirmed our hypothesis, revealing that as Top-K increases, the utility decreases. Interestingly, we discovered that even with small Top-K values, the exposure of the canary data did not exceed the exposure threshold. This observation suggests that the CUSTEXT+ algorithm effectively preserves privacy, even when the number of substitute words is limited. Based on these findings, we selected Top-K = 10.

• **Mahalanobis Mechanism:** Xu et al. [42] introduce a parameter λ to control the degree of regularization in the Mahalanobis norm. This parameter controls the shape of the equidistant contours in the noise distribution used for text perturbation: higher λ values stretch the noise distribution more along directions where word embeddings show the most variation. This stretching increases the likelihood of replacing words in sparse regions of the embedding space while preserving the overall noise scale. To determine the optimal λ value, we conducted experiments with values ranging from 0.1 to 1.0. Our experimental results shown in Figure 5 indicate that the highest utility was achieved when λ was set to 1.0. We also found that canary data remained well-protected across all tested λ

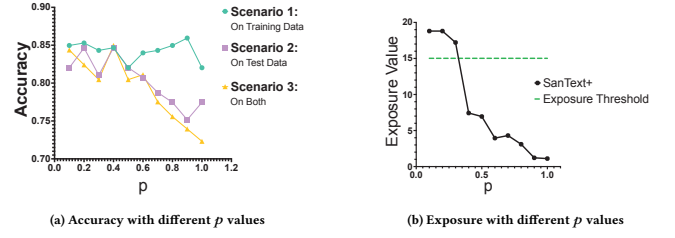


Figure 3: Optimized Parameter Selection for SANTEXT+

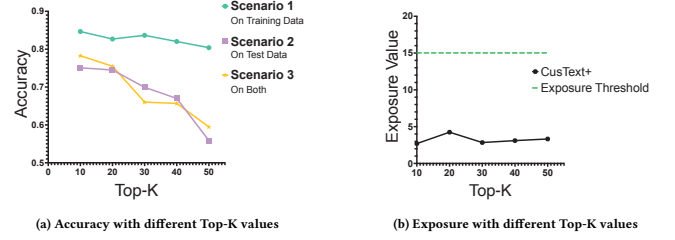


Figure 4: Optimized Parameter Selection for CUSTEXT+

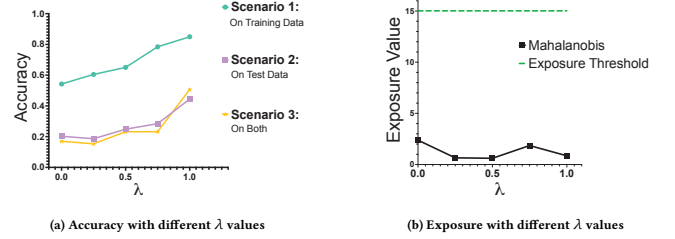


Figure 5: Optimized Parameter Selection for Mahalanobis Mechanism

values, with exposure value consistently staying below the exposure threshold. Based on these observations, we selected $\lambda = 1.0$.

4.4 Utility Results

Having optimized parameters for each privacy-preserving algorithm, we conduct our evaluation of three algorithms across diverse datasets: MTSamples dataset, SST-2, and QNLI in three scenarios shown in Figure 2. We vary the privacy budget ϵ from 1 to 10, allowing us to observe how the algorithms behave under different privacy constraints. To systematically evaluate algorithm performance, we establish an "**acceptable accuracy threshold**" for each dataset. These thresholds serve as benchmarks for assessing the effectiveness of privacy-preserving algorithms under various conditions. We set the acceptable accuracy at **0.80 for MTSamples**, **0.75 for SST-2**, and **0.70 for QNLI**, based on their respective task requirements and complexity. Organizations can adjust these thresholds according to their specific operational needs.

Our utility analysis aims to address three research questions: **RQ1:** What is the minimum privacy budget (ϵ) at which each algorithm can maintain acceptable model accuracy and what factors contribute to the performance differences among these algorithms?

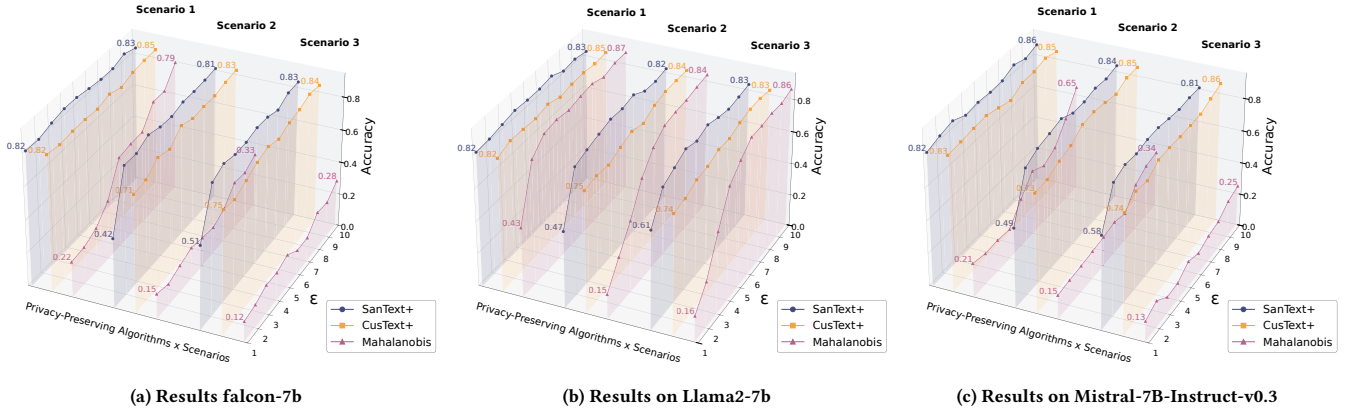


Figure 6: Privacy-Preserving Algorithms under Multiple Scenarios: A Cross-Model Utility Analysis on MTsamples

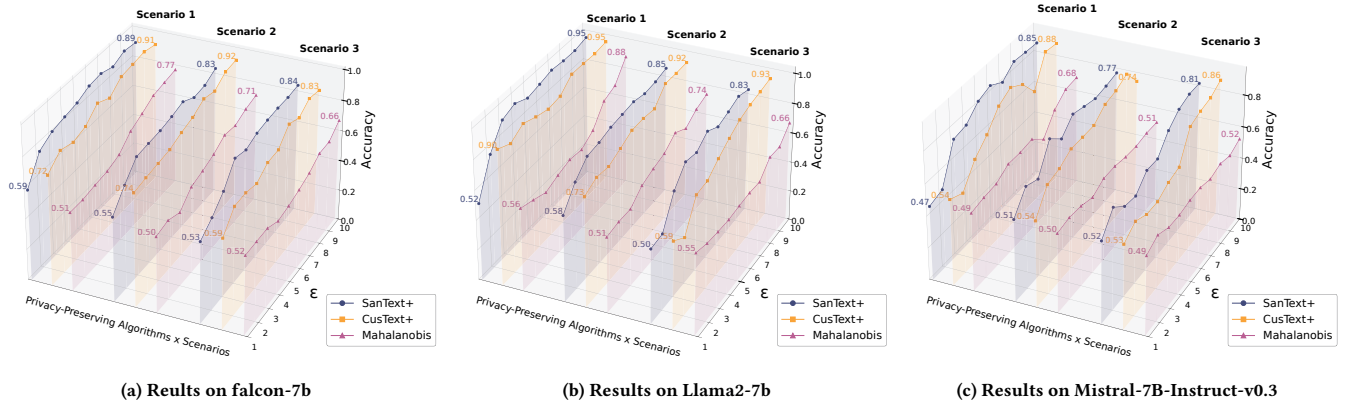


Figure 7: Privacy-Preserving Algorithms under Multiple Scenarios: A Cross-Model Utility Analysis on SST-2 Dataset

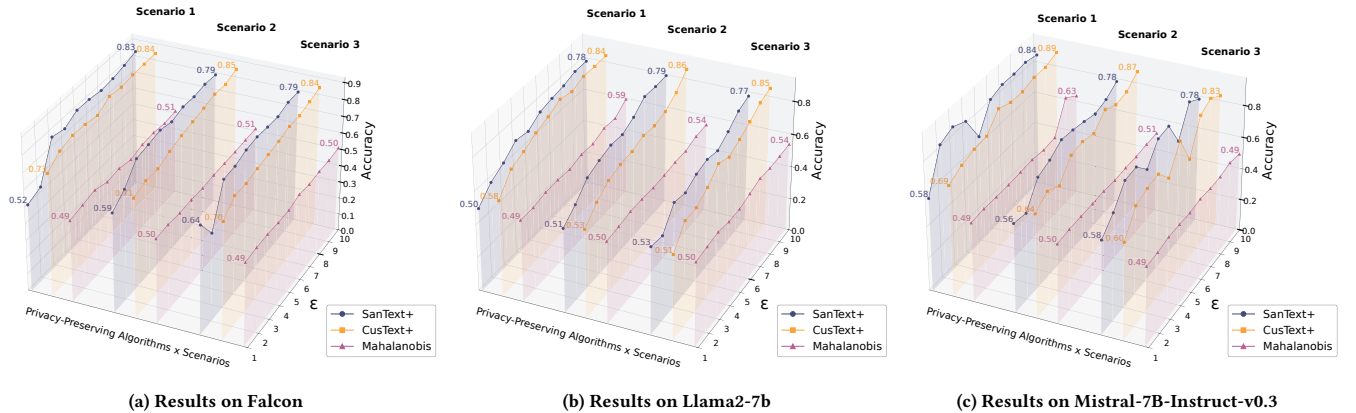


Figure 8: Privacy-Preserving Algorithms under Multiple Scenarios: A Cross-Model Utility Analysis on QNLI Dataset

RQ2: What is the impact of LLM architecture on algorithm performance? **RQ3:** What is the relative effectiveness and suitability of these algorithms across different scenarios?

• **Comparative Analysis of Algorithm Performance:** As shown in Figures 6, 7, and 8, the **Mahalanobis mechanism** achieves acceptable accuracy only in specific conditions: on Llama2-7b with MTsamples dataset, it reaches the threshold at $\epsilon = 3$ in Scenario 1 (accuracy = 0.85) and at $\epsilon = 6$ in Scenarios 2 and 3 (accuracy = 0.80).

and 0.85 respectively). For SST-2, it attains acceptable accuracy (0.75) at $\epsilon = 9$ in Scenario 1 on both Llama2-7b and Falcon-7b. This algorithm fails to reach acceptable accuracy thresholds in all other configurations. Furthermore, the **Mahalanobis mechanism** shows significantly lower performance compared to CusText+ and SANText+, particularly in the QNLI dataset where its utility lags behind other two algorithms by approximately 20 – 30%. While the mechanism shows comparatively favorable accuracy on MTSamples with Llama2-7b, its performance degrades significantly under stricter privacy settings ($\epsilon = 1$), with accuracy dropping approximately 55 – 65% below CusText+ and 25 – 60% below SANText+. This performance gap can be attributed to **three factors**: (1) Its word replacement options are limited to the target dataset, while CusText+ and SANText+ leverage the broader GloVe dataset, allowing more meaningful word substitutions, especially under strict privacy requirements; (2) The Mahalanobis mechanism suffers from the curse of dimensionality when adding noise directly to high-dimensional word embeddings. This leads to unpredictable perturbations that may significantly alter the semantic meaning. In contrast, both SANText+ and CusText+ solve this problem by directly sampling discrete words for replacement; (3) The Mahalanobis mechanism performs especially poorly on the QNLI dataset, where important words are widely scattered throughout the text. This sparse distribution of key terms, combined with the mechanism's difficulty in preserving relationships between words, makes it particularly unsuitable for maintaining the meaning of QNLI sentences.

For other two algorithms, **SANText+** reaches the threshold at ϵ between 1 and 3 for Scenario 1, while requiring ϵ between 3 and 5 for Scenarios 2 and 3 (with Mistral-7B requiring up to $\epsilon = 7$). In comparison, **CusText+** achieves acceptable accuracy at similar budgets (ϵ between 1 and 3) for Scenario 1, but performs better in Scenarios 2 and 3, reaching the threshold at lower budgets (ϵ between 2 and 3), though Mistral-7B still requires up to $\epsilon = 7$. Further analysis reveals that **CusText+** generally shows better performance than **SANText+**, though this performance gap narrows when privacy constraints are more relaxed ($\epsilon \geq 3$). The advantage becomes significantly more evident under strict privacy constraints ($\epsilon = 1$), especially in Scenario 2 and Scenario 3, where **CusText+**'s average accuracy exceeds **SANText+**'s by approximately 32% and 14% respectively. This performance pattern can be attributed to **three reasons**: (1) CusText+ customizes substitutions for each word based on semantic similarity, preserving the coherence of the original sentence. In contrast, SANText+ samples from a wider pool of words, potentially disrupting the semantic integrity of the sentence; (2) CusText+ uses advanced semantic representations, such as Counter-Fitting vectors, to identify relevant substitutions. On the other hand, SANText+ is constrained by the privacy definitions of the Metric Differential Privacy (MLDP) framework, limiting it to comparisons based on Euclidean distance; (3) CusText+ preserves common connecting words, making sentences easier to understand.

• **Comparative Performance Across LLM Architectures**: Our analysis reveals consistent performance patterns for privacy-preserving algorithms across the three LLM architectures. However, the Mahalanobis mechanism exhibits significantly better performance on **Llama2-7b** compared to other models, mainly when applied to the MTSample dataset. This suggests that Llama2-7b shows greater

robustness in handling text modifications introduced by different privacy-preserving algorithms.

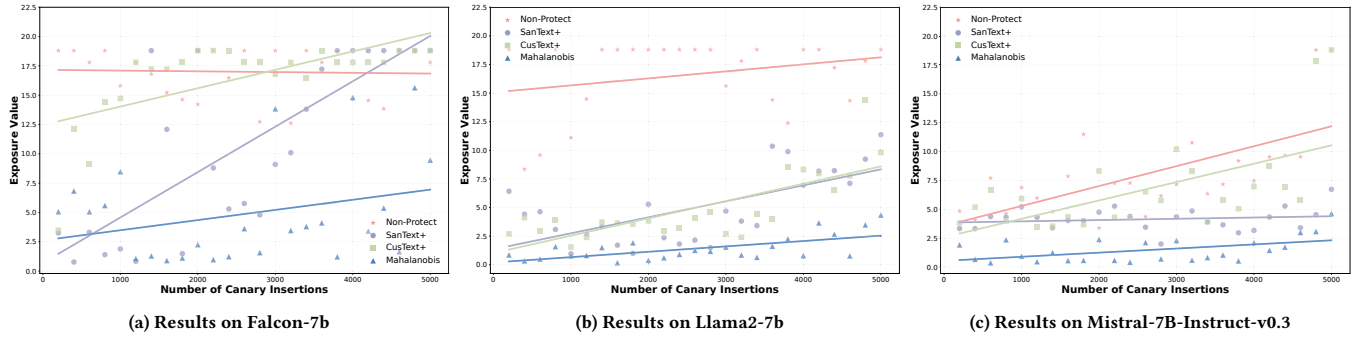
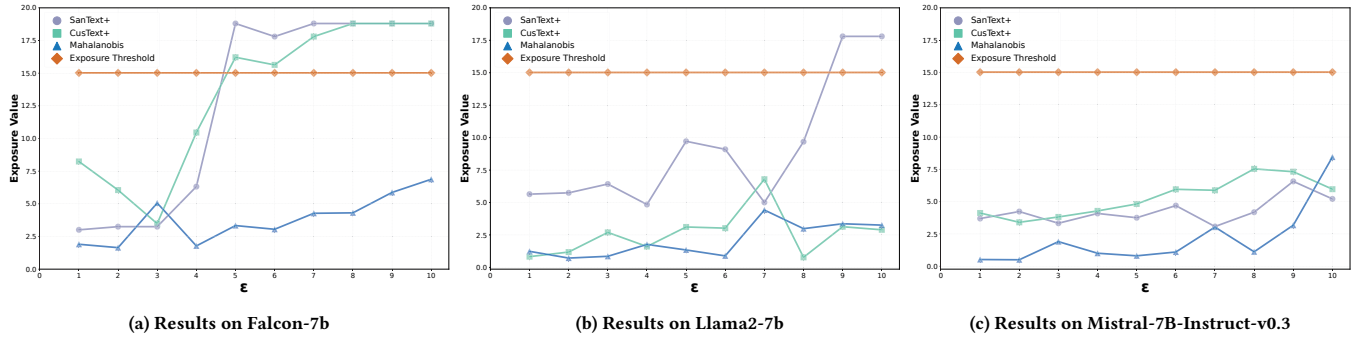
• **Algorithm Performance and Suitability Across Scenarios**: Figures 6, 7, and 8 show that all evaluated algorithms achieve significantly lower performance in Scenarios 2 and 3 compared to Scenario 1. This observation can be attributed to the fact that current algorithms are primarily designed for training datasets. Developing privacy-preserving algorithms tailored to query data is essential for better utility-privacy trade-offs in real-world scenarios.

Furthermore, among the evaluated algorithms, only **SANText+** and **CusText+** are appropriate for protecting individual data points. These methods are based on Metric Local Differential Privacy (MLDP) [7], which is derived from Local Differential Privacy (LDP) [12]. LDP mechanisms enable data owners to sanitize their data locally before releasing it to the server, allowing individual query data points to be processed by **SANText+** or **CusText+** before being sent to the LLM server, making them ideal for scenarios requiring query data protection. In contrast, the Mahalanobis mechanism employs Global Differential Privacy, which is not well-suited for protecting individual data points.

4.5 Privacy Results

Our privacy analysis examines the effectiveness of privacy-preserving algorithms by evaluating privacy levels using exposure metric under varying conditions. Our analysis uses the MTSample dataset with a consistent training duration of 120 epochs across all models and algorithms to ensure a fair comparison. In this section, we address the research question: **RQ**: *To what extent do LLM architectures (Mistral-7B, Llama2-7b, Falcon-7b) affect the measured exposure values of privacy-preserving algorithms under two controlled conditions: (1) fixed privacy budgets (ϵ) with varying canary data insertions; (2) fixed numbers of canary insertions with varying privacy budgets?*

• **Effect of Increasing Canary Data Insertions on Privacy Protection**: Figure 9 shows how canary data exposure varies across different LLMs as the number of canary insertions increases while maintaining $\epsilon = 3$. The Mahalanobis mechanism consistently maintains minimal exposure values, showing only slight increases with additional insertions. In contrast, both **CusText+** and **SANText+** show more pronounced increases in exposure as insertions grow. For **Falcon-7b** without privacy protection, high exposure values persist from 200 to 5000 insertions. While **CusText+** and **SANText+** effectively protect canary data at low insertion counts, their effectiveness diminishes with larger volumes. **CusText+** reaches high exposure values first, followed by **SANText+**, indicating that both algorithms struggle to maintain privacy when handling large volumes of repetitive canary data on Falcon-7b. This suggests the need for either dataset deduplication or algorithm parameter adjustments. **Mistral-7B** demonstrates inherent resilience to the extraction process, reaching high exposure values only at very high insertion counts (≥ 4800) without privacy protection. The privacy-preserving algorithms further enhance this resistance, effectively reducing exposure values. **Llama2-7b**, despite showing high exposure without protection, maintains robust privacy when using any of the privacy-preserving algorithms, even at maximum insertion counts (5000). This indicates exceptional compatibility between Llama2-7b and the algorithms.

Figure 9: Privacy Exposure Analysis under Varying Canary Insertion Rates across different LLMs ($\epsilon = 3$)Figure 10: Privacy Exposure Analysis under Varying Privacy Budgets (ϵ) across Different LLMs (Canary Insertions = 200)

These findings reveal that different LLMs exhibit varying privacy characteristics and algorithm compatibility. Mistral-7B emerges as a practical choice for real-world applications due to its inherent resistance to canary data exposure, while Llama2-7b stands out for its strong performance across all privacy-preserving algorithms.

• **Impact of Increasing Privacy Budget on Canary Data Exposure:** Figure 10 indicates the exposure value when testing with 200 canary insertions across increasing ϵ values. The Mahalanobis mechanism consistently maintains complete privacy protection, as no exposure value exceeds the exposure threshold. In contrast, both CusText+ and SanText+ exhibit varying levels of vulnerability as privacy budgets become less restrictive (higher ϵ values).

The effectiveness of privacy protection varies significantly across LLM architectures. Falcon-7b shows the earliest signs of vulnerability, with canary data exposed beginning at $\epsilon = 5$. Mistral-7B maintains robust privacy protection across all privacy-preserving algorithms, because of its inherent defensive capabilities. On Llama2-7b, CusText+ achieves complete privacy protection across all ϵ values, while SanText+ only shows vulnerability when $\epsilon \geq 9$.

5 Conclusion

This study presents a comprehensive evaluation of data-level privacy-preserving algorithms applied to Large Language Models (LLMs). We assessed three algorithms - SanText+, CusText+, and the Mahalanobis mechanism - across three LLMs (Falcon-7b, Llama2-7b, and Mistral-7B) using diverse datasets. Our analysis reveals that each privacy-preserving algorithm shows distinct strengths and

limitations. Mahalanobis mechanism achieves complete privacy protection but shows limited utility. CusText+ emerges as the optimal choice for maintaining utility while providing robust privacy guarantees under strict privacy constraints. Furthermore, the three LLMs exhibited distinct privacy characteristics: Falcon-7b shows an increased vulnerability to information exposure, particularly with relaxed privacy settings or frequent occurrences of canary data. Mistral-7B demonstrates strong inherent privacy preservation, requiring minimal additional protection. Notably, Llama2-7b exhibits exceptional compatibility with privacy-preserving algorithms, consistently maintaining both effective privacy protection and high utility across different privacy-preserving approaches.

In conclusion, privacy protection in LLMs remains a major challenge for both academia and industry, especially as these models increasingly handle sensitive data in healthcare, finance, and education. Our evaluation framework establishes a foundation for standardizing privacy-preserving algorithms assessments in LLMs, facilitating the development of more sophisticated and LLM-specific privacy-preserving algorithms. Future research could focus on: (1) exploring diverse attack scenarios to enhance robustness evaluation, (2) developing specialized algorithms for query data protection, and (3) Investigating the impact of privacy protection on system performance metrics including inference latency.

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A Parameter Optimizaton Algorithm

Symbols	Description
PPA	Privacy-preserving algorithm
$\mathcal{P}$	The parameter p that affects the utility and degree of privacy protection.
TRA	The training action.
TE	The testing action.
$EP(LLM, D_{TA})$	Calculate the exposure of the LLM using the training dataset.
D_{TA}	Training dataset.
D_{TE}	Testing dataset.
CD	Canary data, which is one piece of private information.
D_{CTA}	The training dataset which includes canary data.
$ETS(LLM, D_{TA}, D_{TE})$	Evaluate the utility of the LLM under three conditions.
$PPL(D)$	Apply the privacy-preserving algorithm to the dataset.
LLM	Large Language Model.
EPV	Exposure threshold value, indicating that canary data is exposed if its exposure exceeds this value.
U	Utility, which could be decided by accuracy.

Table 1: Description of Symbols Used in the Study

Algorithm 1 Parameter Optimization

```

Input: a set of  $\mathcal{P}, CD, D_{TA}, D_{TE}, LLM$ 
for each  $p \in$  set of  $\mathcal{P}$  values do
     $D_{TA, \mathcal{P}} \leftarrow PPL(D_{TA}, \mathcal{P})$ 
     $D_{TE, \mathcal{P}} \leftarrow PPL(D_{TE}, \mathcal{P})$ 
     $U \leftarrow ETS(LLM, D_{TA, \mathcal{P}}, D_{TE, \mathcal{P}})$ 
end for
 $D_{CTA} \leftarrow D_{TA} + CD$ 
 $Valid_{\mathcal{P}} \leftarrow \{\}$ 
for each  $\mathcal{P} \in$  set of  $\mathcal{P}$  do
     $D_{CTA, \mathcal{P}} \leftarrow PPL(D_{CTA}, \mathcal{P})$ 
     $Exposure \leftarrow EP(LLM, D_{CTA, \mathcal{P}})$ 
    if  $Exposure < EPV$  then
         $Valid_{\mathcal{P}}.append(\mathcal{P})$ 
    end if
end for
 $\mathcal{P}_{best} \leftarrow argmax(U \text{ of each } \mathcal{P} \in Valid_{\mathcal{P}})$ 
Output:  $\mathcal{P}_{best}$ 

```

B Resource Cost Test

Resource allocation is critical when implementing Large Language Model (LLM) projects due to their computational intensity. This study evaluates the performance of various GPUs available on Google Colab (A100, V100, L4, and T4) to optimize resource utilization. We assessed the resource costs of fine-tuning Llama2-7b using the MTSamples dataset comprising 1,226 training and 306 testing samples over 120 epochs. During inference, we set the ‘max_new_tokens’ parameter to 15.

The results in Table 2 indicate that the A100 GPU offers the fastest training time but at the highest compute unit cost, while the V100, L4 and T4 GPUs exhibit longer training times with lower costs. This finding highlights the trade-off between computational speed and resource expenditure, allowing organizations to make decisions based on their specific requirements and financial limitations.

Notably, GPU choice minimally impacted testing time, with minor variations attributable to the stochastic nature of Llama2-7b.

Table 2: Resource Cost for Different GPUs

	A100 GPU	V100 GPU	L4 GPU	T4 GPU
Training time	174.02	236.67	422.46	695.49
Testing time	487.37	514.03	503.20	608.45
Compute units per hour	11.77	4.91	4.82	1.84

This suggests organizations could optimize resource allocation by using high-performance GPUs for training and cost-effective GPUs for testing and querying. Furthermore, contrary to expectations, increasing the ‘max_sequence’ value did not significantly affect training time. However, as shown in Table 3, the ‘max_new_tokens’ parameter directly influenced testing time, with higher values increasing computational demands. Organizations should consider these findings when determining output length requirements, particularly for deployed LLMs. For classification tasks, lower ‘max_new_tokens’ values suffice, while chat or advice-generation applications may require higher values to meet functional needs.

Table 3: Time Costs Associated with Different Token Generation Limits

max_new_tokens	5	10	15	20	30	50	100
test time	191.08	318.53	424.84	576.89	708.19	1387.34	2743.24

C Hyperparameters Setting

Llama2-7b			
Dataset	MTSamples	SST-2	QNLI
Epoch	120	50	150
Batch Size	4	4	4
Learning Rate	4e-4	4e-4	4e-4
Mistral-7B			
Dataset	MTSamples	SST-2	QNLI
Epoch	240	120	120
Batch Size	4	4	4
Learning Rate	4e-4	8e-4	4e-4
falcon-7b			
Dataset	MTSamples	SST-2	QNLI
Epoch	360	120	480
Batch Size	4	4	4
Learning Rate	4e-4	4e-4	6e-4

Table 4: Training Hyperparameters

Dataset	MTSamples	SST-2	QNLI
Top_k	10	10	10
Max_new_tokens	15	10	4

Table 5: Testing Hyperparameters